Review of Last Lecture

4.5 Inverse Laplace Transform

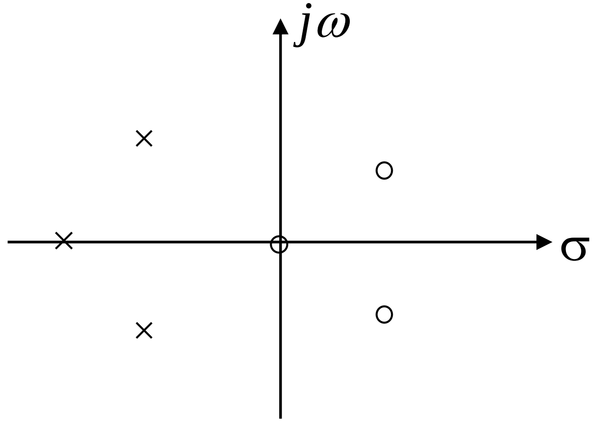
4.6 Analyzing Circuits and S-domain Models via Inverse Laplace Method

4.5.1 Zeros and Poles of Laplace Transform

$$f(t)=\mathcal{L}^{-1}\{F(s)\}=\frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty}F(s)e^{st}ds$$

According to the formula of the inverse Laplace transform, contour integration can be applied to solve it. However, in practice, most Laplace transforms are in the form of **rational fractions**:

$$F(s)=\frac{N(s)}{D(s)}=\frac{\sum_{i=0}^{m}b_i s^i}{\sum_{i=0}^{n}a_i s^i}=\frac{\prod_{i=1}^{m}(s-z_i)}{\prod_{i=1}^{n}(s-p_i)}$$



Zero-pole Distribution Diagram

The roots $\mathbf{z_i}$ of the numerator polynomial $N(s)=0$ are called the **zeros** of ***F(s)***;
the roots $\boldsymbol{p_i}$ of the denominator polynomial $D(s)=0$ are called the **poles** of ***F(s)***

4.5.2 Finding the Inverse Laplace Transform

When analyzing systems using the Laplace transform method, the final step is to perform the **inverse Laplace transform** on the image function.

1. Partial Fraction Method (Heaviside Method)

(1) Case of simple real poles $(p_1 \cdots p_n)$

When $m < n$, $F(s) \overset{Decomposition}{==} \dfrac{k_1}{s-p_1} + \cdots + \dfrac{k_n}{s-p_n}$ **(**Proper Fraction**)**

where $k_i = (s-p_i)F(s)\big|_{s=p_i}$ (Residue) (See the textbook for the proof)

It is known from the properties of Laplace transform：

$$Ae^{-\alpha t}u(t) \overset{LT}{\longleftrightarrow} \frac{A}{s+\alpha}$$

$$\boxed{F(s) \xrightarrow{L^{-1}} f(t) = \left(k_1 e^{p_1 t} + \cdots + k_n e^{p_n t}\right)u(t)}$$

(2) The case where poles include conjugate complex roots $(p_{1,2} = -\alpha \pm j\beta)$

$$F(s) = \frac{A(s)}{D(s)\left[(s+\alpha)^2 + \beta^2\right]}$$

$$= \frac{A(s)}{D(s)(s+\alpha - j\beta)(s+\alpha + j\beta)}$$

where $D(s)$ is the remaining part of the denominator after removing the conjugate complex roots.

Set $F_1(s) = \dfrac{A(s)}{D(s)}$

then $F(s) = \dfrac{F_1(s)}{(s+\alpha - j\beta)(s+\alpha + j\beta)}$

$$\overset{Decomposition}{==} \frac{K_1}{s+\alpha - j\beta} + \frac{K_2}{s+\alpha + j\beta} + \dots$$

where $K_{1,2} = \dfrac{F_1(-\alpha \pm j\beta)}{\pm 2j\beta}$ (residue)

K_1、K_2 have a conjugate relationship: $K_{1,2} = A \pm jB$

Thus, the inverse transform of the part related to the conjugate complex roots

is $\xrightarrow{\text{LT}^{-1}}$

$$f_c(t) = e^{-\alpha t}(K_1 e^{j\beta t} + K_1^* e^{-j\beta t})u(t)$$

$$= e^{-\alpha t}\{(A + jB)[\cos(\beta t) + j\sin(\beta t)] + (A - jB)[\cos(\beta t) - j\sin(\beta t)]\}u(t)$$

$$= 2e^{-\alpha t}\left[A\cos(\beta t) - B\sin(\beta t)\right]u(t)$$

α--attenuation factor; β--oscillation frequency

(3) The case where poles include multiple roots(a k-fold p_1, simple roots $p_2 \ldots p_n$)

$$F(s)=\frac{A(s)}{D(s)}=\frac{A(s)}{(s-p_1)^k D_1(s)}$$

$$=\frac{K_{11}}{(s-p_1)^k}+\frac{K_{12}}{(s-p_1)^{k-1}}+\cdots+\frac{K_{1k}}{(s-p_1)}+\sum_{i=2}^{n}\frac{K_i}{s-p_i}$$

$$=\sum_{i=1}^{k}\frac{K_{1i}}{(s-p_1)^{k-i+1}}+\sum_{i=2}^{n}\frac{K_i}{s-p_i}$$

To find K_i and K_{11}, the method is the same as the first case:

$$K_{11}=(s-p_1)^k F(s)\Big|_{s=p_1}$$

To find other coefficients,

Let $F_1(s)=(s-p_1)^k \cdot F(s)$

when $i=2$, $\quad K_{12}=\dfrac{\mathrm{d}}{\mathrm{d}\,s}F_1(s)\Big|_{s=p_1}$

when $i=3$, $\quad K_{13}=\dfrac{1}{2}\dfrac{\mathrm{d}^2}{\mathrm{d}\,s^2}F_1(s)\Big|_{s=p_1}$

$$K_{1i} = \frac{1}{(i-1)!} \frac{\mathrm{d}^{i-1}}{\mathrm{d}\, s^{i-1}} F_1(s) \Big|_{s=p_1} \qquad (i = 1, 2, \cdots k)$$

$$F_1(s) = (s - p_1)^k \cdot F(s)$$

The inverse transform of the multiple root part $\xrightarrow{\mathrm{LT}^{-1}}$:

$$f_c(t) = e^{p_1 t} \left[\frac{K_{11}}{(k-1)!} t^{k-1} + \frac{K_{12}}{(k-2)!} t^{k-2} \cdots + \frac{K_{1i}}{(k-i)!} t^{k-i} \cdots + K_{1k} \right] u(t)$$

单选题 1分

设置

Given $F(s)=\left(\frac{1-e^{-s}}{s}\right)^2$, find its Inverse Laplace Transform $f(t)$.

A $f(t)=tu(t)-2(t-1)u(t-1)+(t-2)u(t-2)$

B $f(t)=(t-1)u(t-1)+(t-2)u(t-2)$

C $f(t)=tu(t)-(t-1)u(t-1)+(t-2)u(t-2)$

D $f(t)=tu(t)-(t-2)u(t-2)$

提交

Given $F(s)=\left(\frac{1-\mathrm{e}^{-s}}{s}\right)^2$, find its Inverse Laplace Transform $f(\mathrm{t})$.

Solution: $F(s)=\frac{1-2\mathrm{e}^{-s}+\mathrm{e}^{-2s}}{s^2}=\frac{1}{s^2}-\frac{2}{s^2}\mathrm{e}^{-s}+\frac{1}{s^2}\mathrm{e}^{-2s}$

$$tu(t)\leftrightarrow\frac{1}{s^2}$$

Using the Time-Shifting Property:

$$\therefore\ f(t)=tu(t)-2(t-1)u(t-1)+(t-2)u(t-2)$$

Even if ***F*(*s*)** is not a proper fraction, i.e. ***n*<*m*,** it can also be expressed as the sum of a polynomial in ***s*** and a **proper fraction**:

Example: $$F(s)=\frac{s^3+4s^2+6s+5}{s^2+3s+2}=s+1+\frac{s+3}{s^2+3s+2}$$

In the above equation, the inverse Laplace transform of ***s*+1** is equal to : $\delta'(t)+\delta(t)$

For the inverse transform of a **rational proper fraction** in Laplace transform, solve it using the **partial fraction method**.

If ***F*(*s*)** is an **irrational function** containing $e^{-\alpha s}$, the term $e^{-\alpha s}$ does not participate in partial fraction decomposition, and the solution utilizes the **Time-Shifting Property.**

Example: $$\frac{\mathrm{e}^{-2s}}{s^2+3s+2}=F_1(s)\mathrm{e}^{-2s} \qquad F_1(s)=\frac{1}{s+1}+\frac{-1}{s+2}$$

So $f_1(t)=L^{-1}\left[F_1(s)\right]=\left(\mathrm{e}^{-t}-\mathrm{e}^{-2t}\right)u(t)$

So $f(t)=f_1(t-2)=\left[\mathrm{e}^{-(t-2)}-\mathrm{e}^{-2(t-2)}\right]u(t-2)$

4.6.1 Solving Differential Equations Using Laplace Transform

Steps for Solving Differential Equations Using Laplace Transform:

(1) Take the Laplace transform of the differential equation, paying attention to applying the differentiation property;

(2)Solve the algebraic equation in the s-domain to find the Laplace transform of the response;

(3) Find the inverse Laplace transform, i.e., find the time-domain expression of the response.

4.6.2 Solving Circuits Using Laplace Transform

s-domain Circuit Analysis Method

1. Replace each component in the network with its **s-domain model**;
Write the signal source as a transformed expression ($\frac{E}{s}$ or $\frac{E}{sR}$, $U_s(s)$ or $I_s(s)$)

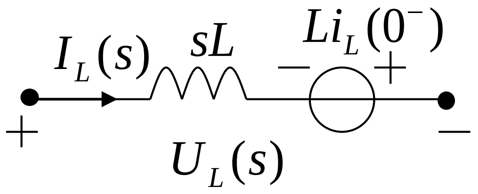
2. Analyze the constructed s-domain model diagram using KVL and KCL to obtain the required transformed system equations.
(The numerical operations performed are algebraic relationships, similar to resistive networks; Thevenin's theorem and Norton's theorem are both applicable; It is suitable for network analysis with many nodes or loops.)

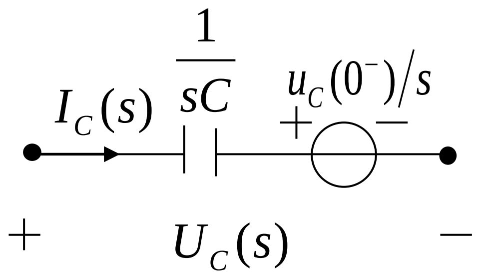
3. Find the **Laplace transform** of the response.

4. Find the **inverse Laplace transform**, i.e., find the time-domain expression of the response.

4.6 Analyzing Circuits and S-domain Models via Inverse Laplace Method

1) Loop Analysis (KVL)

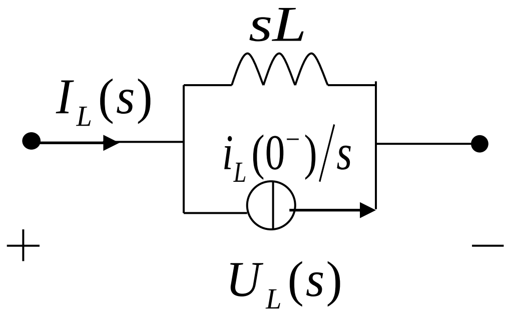


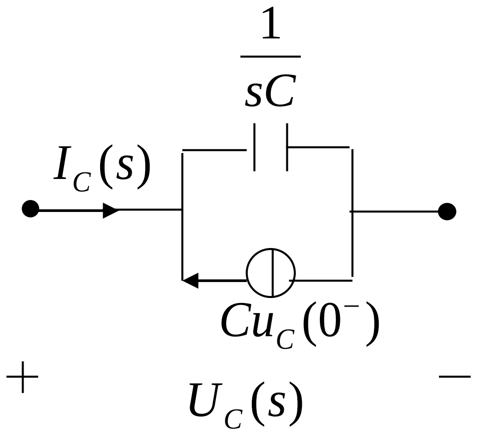


$$U_L(s) = sLI_L(s) - Li_L(0^-)$$

$$U_C(s) = \frac{1}{sC} I_C(s) + \frac{1}{s} u_c(0^-)$$

2) Node Analysis (KCL)





$$I_L(s) = \frac{1}{sL} U_L(s) + \frac{1}{s} i_L(0^-)$$

$$I_C(s) = sCU_C(s) - Cu_c(0^-)$$

Contents of This Lecture

4.7 System Function (Network Function)

4.8 Time-domain Characteristics from System Function Zero-pole Distribution

4.9 Frequency Response from System Function Zero-pole Distribution

Learning Objectives of This Lecture

1. Proficiently grasp the interrelationship among **system function**, **system differential equation**, and **system simulation block diagram** (**Important**);
2. Be able to qualitatively determine the characteristics of **impulse response** using the zero-pole distribution of **system function**;
3. Proficiently use **system function** to analyze the time-domain characteristics of **zero-state response**;
4. Proficiently use the zero-pole distribution of **system function** to roughly sketch the **frequency response characteristic** curve (**Important**).

4.7.1 Definition of System Function

If a time signal acts on a system and its output is still this time signal, only the **amplitude and phase are changed**, this time signal is called the **characteristic signal** of the system. The function that characterizes the changed amplitude and phase is called the **eigenvalue** or **system function** of the system.

The signal $e^{j\omega t}$ is the **characteristic signal** of the system, and the function $H(\omega)$ (the Fourier transform of the system's unit impulse response **$h(t)$** is the **system function**, also known as the **frequency response** of the system.

$$e^{j\omega t} \longrightarrow \boxed{h(t)} \longrightarrow y(t) = h(t) * e^{j\omega t} = H(\omega)e^{j\omega t} = |H(\omega)|e^{j[\omega t+\varphi(\omega)]}$$

In fact, the **exponential signal e^{st}** is also a **characteristic signal** of the system.

$$e^{st} \longrightarrow \boxed{h(t)} \longrightarrow y(t) = h(t) * e^{st} = H(s)e^{st}$$

$$y(t) = h(t) * e^{st} = \int_{0_-}^{\infty} h(\tau)e^{s(t-\tau)}d\tau = e^{st}\int_{0_-}^{\infty} h(\tau)e^{-s\tau}d\tau = H(s)e^{st}$$

4.7 System Function

$H(s)=\int_{0^-}^{\infty} h(t)e^{-st}dt$ is the Laplace transform of the unit impulse response of a **causal system**, also known as the **system function**.

For a continuous-time signal ***x*(*t*)**, acting on a linear time-invariant system with unit impulse response ***h*(*t*)**, its **zero-state response** $\boldsymbol{y_{zs}(t)}$ is equal to the **convolution** of the input signal and the unit impulse response:

$$x(t) \longrightarrow \boxed{h(t)} \longrightarrow y_{zs}(t)=h(t)*x(t)$$

By the **time-domain convolution theorem**, the Laplace transform of the zero-state response is equal to the **product** of the Laplace transform of the input signal and the Laplace transform of the unit impulse response:

$$X(s) \longrightarrow \boxed{H(s)} \longrightarrow Y_{zs}(s)=H(s)X(s)$$

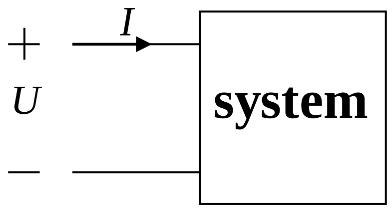
The **system function** is equal to the **ratio** of the Laplace transform of the zero-state response to the Laplace transform of the input signal:

$$\boxed{H(s)=Y_{zs}(s)/X(s)}$$

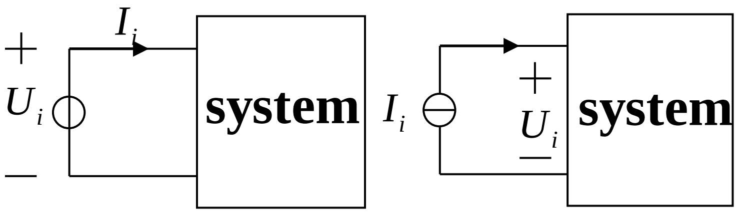
4.7.2 Driving Point Function and Transfer Function

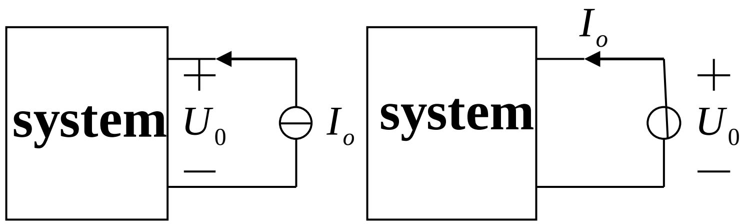
1. Driving Point Function

When the input and output of the system are taken from the same port, the system function is called the **driving point function**.



Depending on its dimension, the driving point function can also be called **impedance function** or **admittance function**. For example, the input or output impedance (ratio of voltage to current) of an amplifier, input or output admittance (ratio of current to voltage), etc.





Impedance: $Z_i(s) = \dfrac{U_i(s)}{I_i(s)}$ $\quad Z_o(s) = \dfrac{U_o(s)}{I_o(s)}$

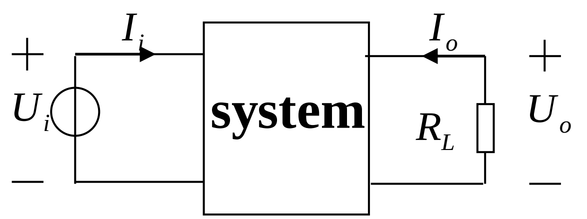
Admittance: $Y_i(s) = \dfrac{I_i(s)}{U_i(s)}$ $\quad Y_o(s) = \dfrac{I_o(s)}{U_o(s)}$

2. Transfer (Transmission) Function

When the input and output of the system are taken from different ports, the system function is called the **transfer function** or **transmission function**.

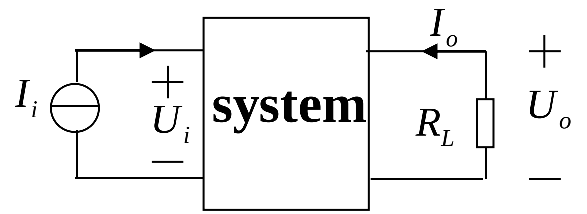
Depending on its dimension, the transfer function can be called **transfer impedance**, **transfer admittance**, **voltage transmission ratio**, or **current transmission ratio**, etc. For example, the voltage or current gain of an amplifier.

Transfer Impedance: $Z_{21}(s)=\dfrac{U_o(s)}{I_i(s)}$



Transfer Admittance: $Y_{21}(s)=\dfrac{I_o(s)}{U_i(s)}$

Voltage Transmission Ratio: $K_v(s)=\dfrac{U_o(s)}{U_i(s)}$



Current Transmission Ratio: $K_i(s)=\dfrac{I_o(s)}{I_i(s)}$

4.7.3 Solution of System Function

Since the system function is the ratio of the Laplace transform of the system's zero-state response to that of the excitation, **its solution is independent of the system's initial conditions**.

Example 4-23: Given the system's differential equation

$$\frac{d^2y(t)}{dt^2}+3\frac{dy(t)}{dt}+2y(t)=\frac{dx(t)}{dt}+3x(t)$$

find the system function $\boldsymbol{H(s)}$ and the system's unit impulse response $\boldsymbol{h(t)}$.

Solution: Take the Laplace transform of the above equation. When using the differential property, **the initial conditions are zero.**

$$(s^2+3s+2)Y(s)=(s+3)X(s)$$

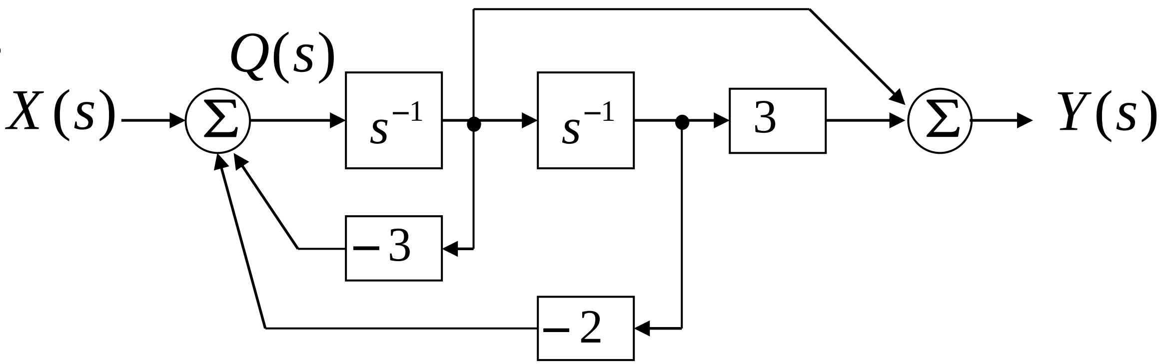
Thus, the system function is

$$H(s)=\frac{Y(s)}{X(s)}=\frac{s+3}{s^2+3s+2}=\frac{s+3}{(s+1)(s+2)}=\frac{2}{s+1}-\frac{1}{s+2}$$

The impulse response of the system is:

$$h(t)=(2e^{-t}-e^{-2t})u(t)$$

Example 4-24: Given the simulation block diagram of a system as follows, find the system function ***H*(*s*)**.



Solution: Let ***Q*(*s*)** be the output of the previous adder. Then:

$$Q(s) = X(s) - 3s^{-1}Q(s) - 2s^{-2}Q(s)$$

$$X(s) = (1 + 3s^{-1} + 2s^{-2})Q(s)$$

The output ***Y*(*s*)** of the latter adder is equal to:

$$Y(s) = s^{-1}Q(s) + 3s^{-2}Q(s) = (s^{-1} + 3s^{-2})Q(s)$$

thus,

$$H(s) = \frac{Y(s)}{X(s)} = \frac{s^{-1} + 3s^{-2}}{1 + 3s^{-1} + 2s^{-2}} = \frac{s+3}{s^2 + 3s + 2}$$

4.8.1 Zero-pole Distribution of System Function and Waveform of Unit Impulse Response

The unit impulse response of a system is the inverse Laplace transform of the system function: $h(t) = L^{-1}\{H(s)\}$

Generally, the system function is a rational fraction, so:

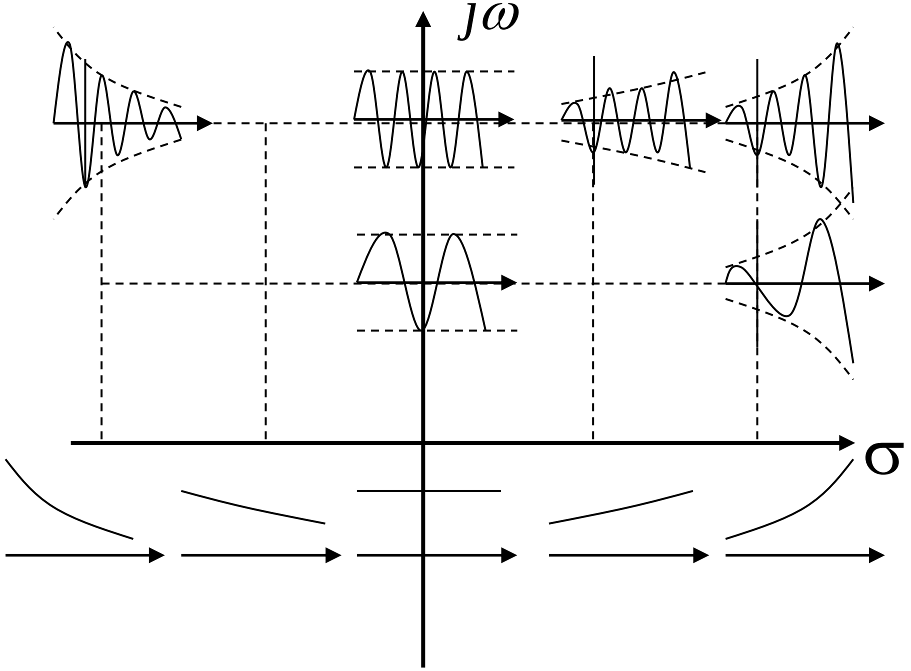
$$h(t) = L^{-1}\{\frac{N(s)}{D(s)}\} = L^{-1}\{\frac{A\prod_{k=1}^{M}(s-z_k)}{\prod_{k=1}^{N}(s-p_k)}\} = L^{-1}\{\sum_{k=1}^{N}\frac{A_k}{s-p_l}\} = \sum_{k=1}^{N} A_k e^{p_k t}u(t)$$

In the above equation, $\boldsymbol{z_k}$ and $\boldsymbol{p_k}$ are the zeros and poles of the system function, respectively. The different positions of zeros and poles in the s-plane have different effects on the unit impulse response.

The **s**-plane is a complex plane, where the ***σ***-axis is the real axis and the imaginary axis is denoted by ***jω***. Different points ***s*** on the plane correspond to different time signals e^{st}.

$$e^{st}=e^{(\sigma+j\omega)t}=e^{\sigma t}\cdot e^{j\omega t}=e^{\sigma t}(\cos\omega t+j\sin\omega t)$$

The signal e^{st} represents a sine-cosine signal with amplitude changing exponentially. **The farther a point is from the real axis, the higher the frequency of the signal change; the farther a point is from the imaginary axis, the faster the signal amplitude grows or decays.**



The points on the real axis represent signals with monotonic exponential changes;

points on the imaginary axis represent sine-cosine signals with constant-amplitude oscillations.

First-order Poles

$H(s)=\dfrac{1}{s}$, $p_1=0$ at the origin, $h(t)=L^{-1}[H(s)]=u(t)$

$H(s)=\dfrac{1}{s+\alpha}$, $p_1=-\alpha$

$\alpha>0$, on the left real axis, $h(t)=\mathrm{e}^{-\alpha t}u(t)$, exponential decay

$\alpha<0$, on the right real axis, $h(t)=\mathrm{e}^{-\alpha t}u(t)$, exponential growth

---

$H(s)=\dfrac{\omega}{s^2+\omega^2}$, $p_{1,2}=\pm\mathrm{j}\omega$, on the imaginary axis

$h(t)=\sin(\omega t)u(t)$, constant – amplitude oscillation

$H(s)=\dfrac{\omega}{(s+\alpha)^2+\omega^2}$, $p_{1,2}=-\alpha\pm\mathrm{j}\omega$ $h(t)=e^{-\alpha t}\sin\omega t u(t)$

When $\alpha>0$, poles in the left half-plane, damped oscillation;

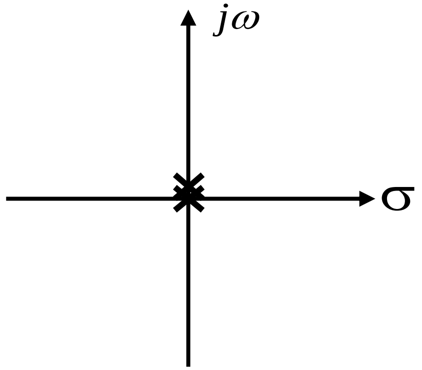
When $\alpha<0$, poles in the right half-plane, growing oscillation

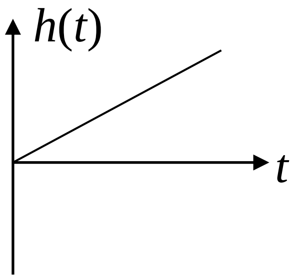
Second-order Poles

$$H(s)=\frac{1}{s^2},\ \text{pole at the origin,}$$

$$h(t)=tu(t)$$

$$t\to\infty, h(t)\to\infty$$

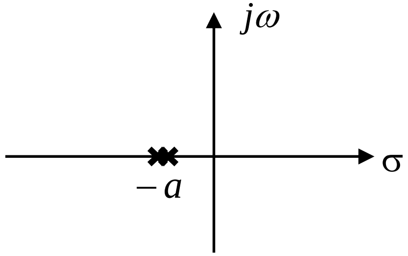


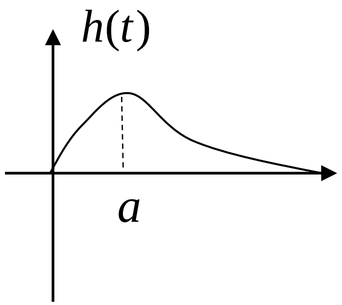


$$H(s)=\frac{1}{(s+\alpha)^2},\ \text{pole on the real axis,}$$

$$h(t)=t\,\mathrm{e}^{-\alpha t}u(t),\alpha>0$$

$$t\to\infty, h(t)\to 0$$

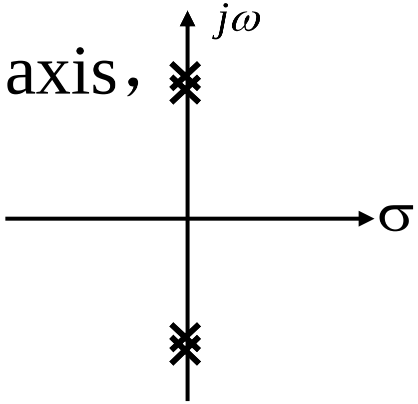


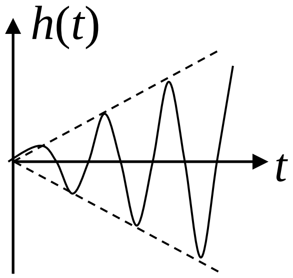


$$H(s)=\frac{2\omega s}{(s^2+\omega^2)^2},\ \text{pole on the imaginary axis,}$$

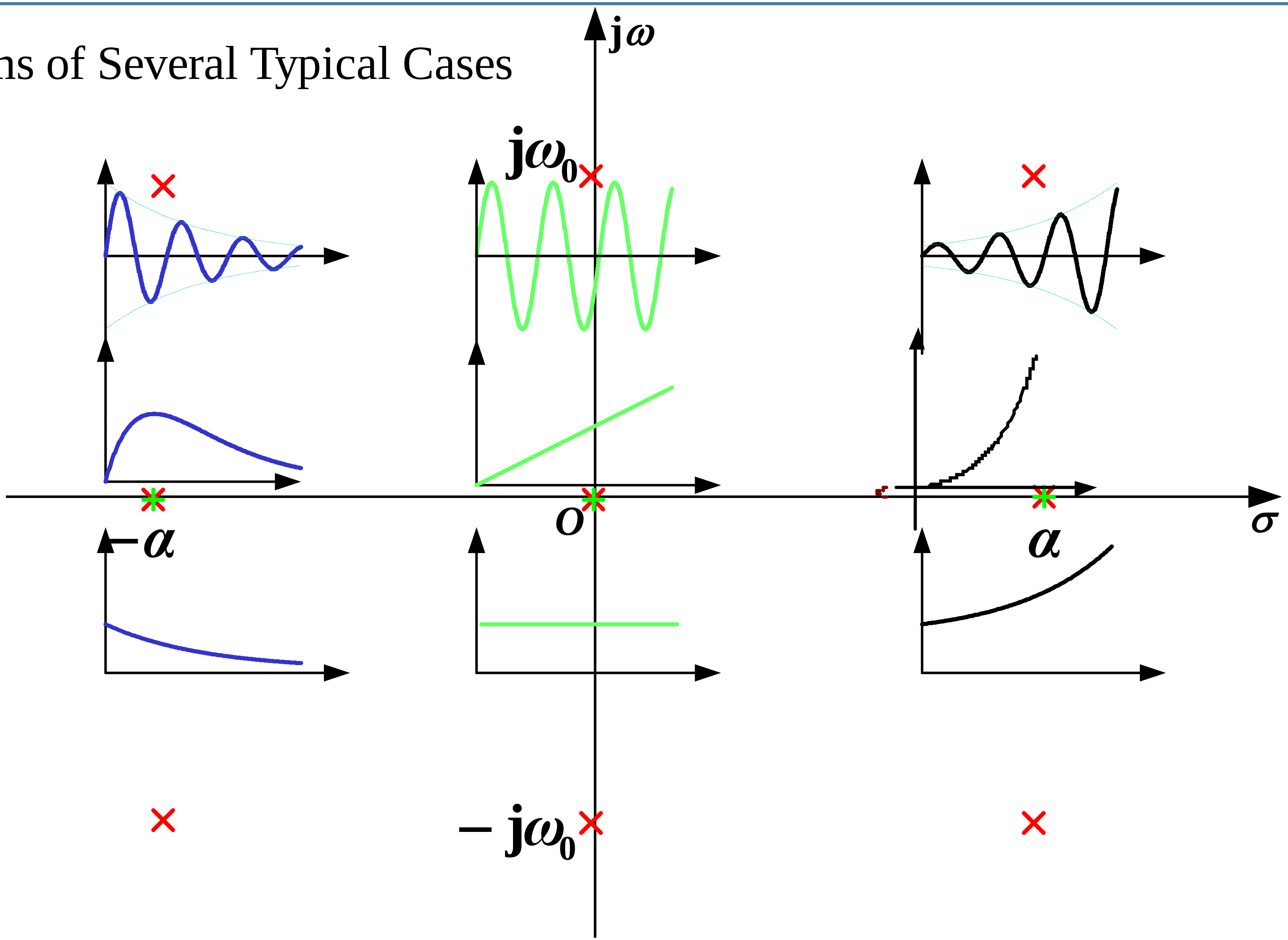
$$h(t)=t\sin\omega t u(t)$$

$t\to\infty, h(t)$ exhibits growing oscillation





Diagrams of Several Typical Cases



Correspondence between Poles of **$H(s)$** and Waveform Characteristics of **$h(t)$**:

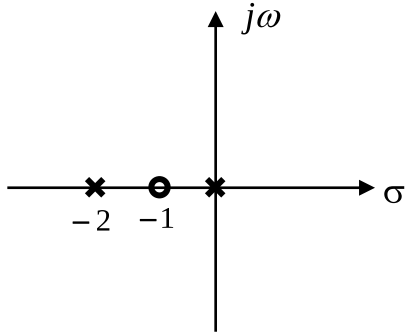
Poles on the left half-plane of the s-plane	**$h(t)$ exhibits a decaying form**
Poles on the right half-plane of the s-plane	**$h(t)$** exhibits a growing form
Poles on the imaginary axis of the s-plane (excluding the origin)	**$h(t)$** has constant-amplitude oscillation (first-order poles) or growing oscillation (multi-order poles)
Poles on the real axis of the s-plane (excluding the origin)	**$h(t)$** exhibits exponential-related changes (first-order or multi-order poles)
Poles at the origin of the s-plane	**$h(t)$** is equal to **$u(t)$**(first-order pole) or grows as t^n (multi-order poles)

Example 4-25: Given the following system functions, sketch their zero-pole plots, and roughly draw the waveforms of their impulse responses.

$$H_1(s)=\frac{s+1}{s(s+2)} \qquad H_2(s)=\frac{s+1}{s^2+2s+2} \qquad H_3(s)=\frac{1}{(s+1)^2}$$

Solution:

The zero-pole plot of $\boldsymbol{H_1(s)}$:

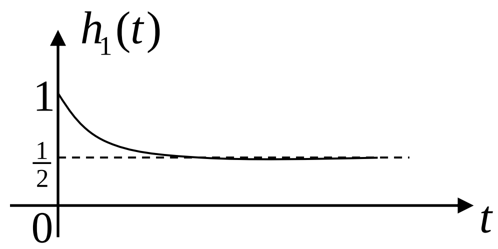


As

$$H_1(s)=\frac{s+1}{s(s+2)}=\frac{1}{2s}+\frac{1}{2(s+2)}$$

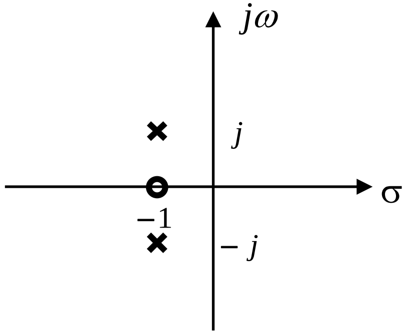
Thus,

$$h_1(t)=\frac{1}{2}(1+e^{-2t})u(t)$$

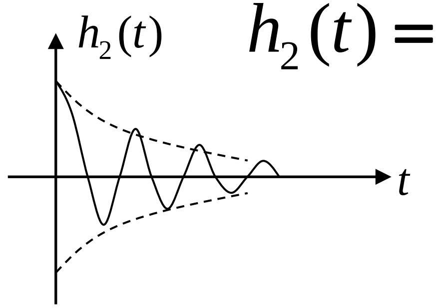


The zero-pole plot of **$H_2(s)$**:

$$H_2(s)=\frac{s+1}{s^2+2s+2}=\frac{s+1}{(s+1)^2+1}$$

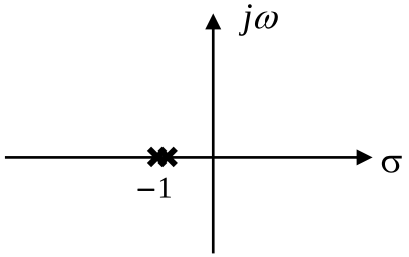


$$h_2(t)=e^{-t}\cos t u(t)$$

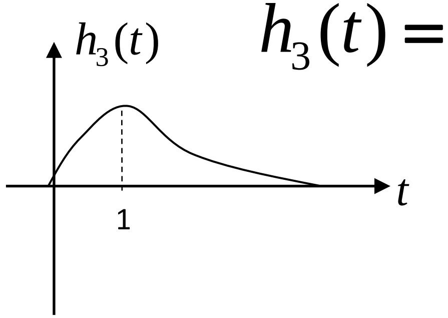


The zero-pole plot of **$H_3(s)$**:

$$H_3(s)=\frac{1}{(s+1)^2}$$



$$h_3(t)=te^{-t}u(t)$$



It can be seen that if a pole is a multi-order pole, its corresponding time-domain waveform is different from that of a first-order pole.

4.8.2 Zero-pole Distribution of System Function & Excitation and Response Components

The **poles** of the system function and the excitation jointly determine the waveform of the response. $Y(s) = X(s)H(s)$

Example 4-26: Let the system function be $H(s) = \frac{s+3}{s^2+3s+2}$, the system excitation be $x(t) = u(t)$, and the initial state be zero. Find the zero-state response of the system, and identify the free response and forced response.

Solution: The Laplace transform of the zero-state response of the system is

$$Y(s) = X(s)H(s) = \frac{1}{s} \cdot \frac{s+3}{s^2+3s+2} = \frac{s+3}{s(s+1)(s+2)} = \frac{\frac{3}{2}}{s} - \frac{2}{s+1} + \frac{\frac{1}{2}}{s+2}$$

Therefore,

the zero-state response of the system is $y(t) = \frac{1}{2}(3 - 4e^{-t} + e^{-2t})u(t)$

where, the free response is $y_h(t) = (\frac{1}{2}e^{-2t} - 2e^{-t})u(t)$, and the forced response is $y_p(t) = \frac{3}{2}u(t)$

The poles of the system function correspond to the free response components; the poles of the excitation correspond to the forced response components.

Example 4-27: Given the differential equation of an LTI system:

$$r''(t) + 2r'(t) + r(t) = e'(t)$$

Find the zero-state response $r_{zs}(t)$ if the excitation is $e(t) = u(t)$.

Solution: Take the Laplace transform of both sides of the differential equation:

$$(s^2 + 2s + 1)R(s) = sE(s)$$

$$H(s) = \frac{R(s)}{E(s)} = \frac{s}{s^2 + 2s + 1} = \frac{s^{-1}}{1 + 2s^{-1} + s^{-2}}$$

Find the Laplace transform of the excitation: $E(s) = \frac{1}{s}$

zero-pole cancellation

$$R_{zs}(s) = H(s)E(s) = \frac{sE(s)}{s^2 + 2s + 1} = \frac{s}{s(s^2 + 2s + 1)} = \frac{1}{s^2 + 2s + 1} = \frac{1}{(s+1)^2}$$

Using the Laplace transform of $tu(t)$ (which is $1/s^2$) and the frequency-shifting property, find the inverse Laplace transform of $R_{zs}(s)$:

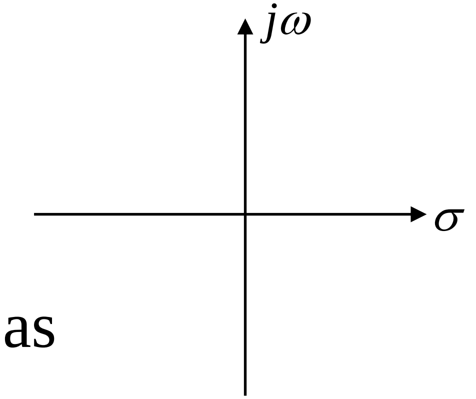
$$r_{zs}(t) = te^{-t}u(t)$$

Natural response only; no forced response due to zero-pole cancellation

4.9 Frequency Response from System Function Zero-pole Distribution

4.9.1 $H(s)$ and $H(\omega)$

The Laplace transform is an extension of the Fourier transform from the real frequency domain **ω** to the complex frequency domain **s**， and the Fourier transform is a special case of the Laplace transform on the imaginary axis of the **s**-plane. **Generally speaking,** $H(\omega) = H(s)\big|_{s=j\omega}$



4.9.2 Zero-pole Distribution of $H(s)$ and $H(\omega)$

Since ***H*(*s*)** is generally a rational fraction, it can be expressed as

$$H(s) = \frac{N(s)}{D(s)} = \frac{G\prod_{k=1}^{M}(s - z_k)}{\prod_{k=1}^{N}(s - p_k)}$$

***G*—constant**

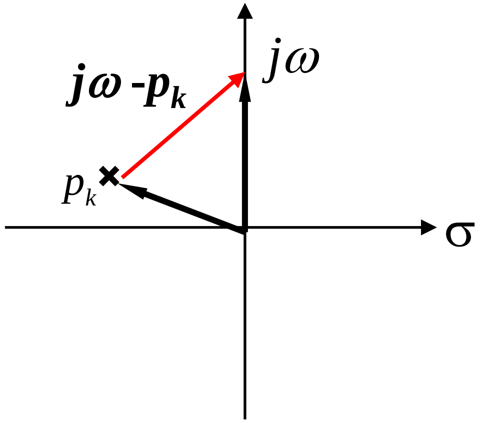
4.9 Frequency Response from System Function Zero-pole Distribution

The frequency response of a system can be **generally** expressed as:

$$H(\omega)=H(s)\big|_{s=j\omega}=\left.\frac{G\prod_{k=1}^{M}(s-z_k)}{\prod_{k=1}^{N}(s-p_k)}\right|_{s=j\omega}=\frac{G\prod_{k=1}^{M}(j\omega-z_k)}{\prod_{k=1}^{N}(j\omega-p_k)}$$

In the above equation, each factor in the numerator and denominator represents a vector on the **s-plane** pointing to the ***jω*-axis.**

The factor **($j\omega$ -p_k)** represents the difference vector between the vector ***jω*** (varying along the imaginary axis) and the vector from the origin to p_k.



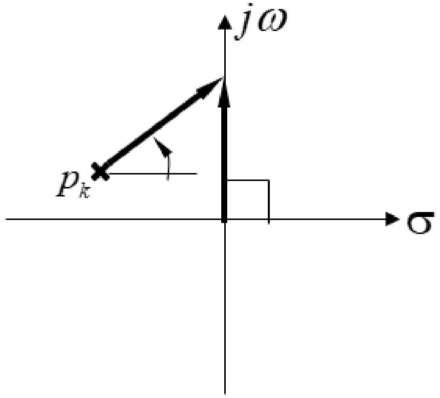
The vector corresponding to the factor in the numerator is called the **zero vector**;

the vector corresponding to the factor in the denominator is called the **pole vector**.

Each vector has its magnitude and phase, so:

$$H(\omega)=\frac{G\prod_{k=1}^{M}(j\omega-z_k)}{\prod_{k=1}^{N}(j\omega-p_k)}=\frac{G\prod_{k=1}^{M}B_k e^{j\beta_k}}{\prod_{k=1}^{N}A_k e^{j\theta_k}}=\frac{G\prod_{k=1}^{M}B_k}{\prod_{k=1}^{N}A_k}e^{j\left(\sum_{k=1}^{M}\beta_k-\sum_{k=1}^{N}\theta_k\right)}$$

$$=\left|H(\omega)\right|e^{j\varphi(\omega)}$$

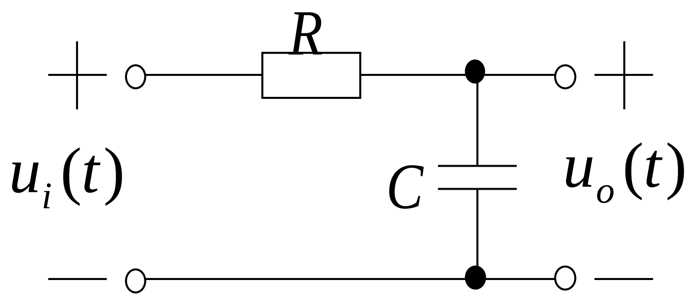


$$\left|H(\omega)\right|=\frac{G\prod_{k=1}^{M}B_k}{\prod_{k=1}^{N}A_k}\qquad \varphi(\omega)=\sum_{k=1}^{M}\beta_k-\sum_{k=1}^{N}\theta_k$$

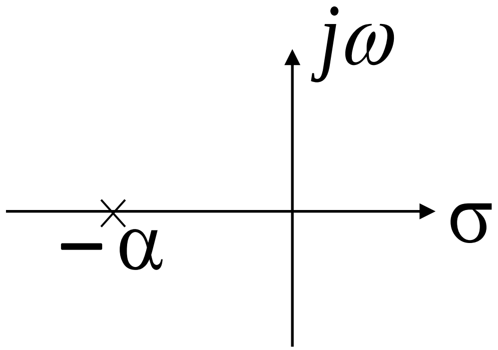
The **magnitude-frequency response** of the system is **the ratio of the product of the magnitudes of zero vectors to the product of the magnitudes of pole vectors**.

The **phase-frequency response** of the system is **the sum of the phases of zero vectors minus the sum of the phases of pole vectors**.

Example 4-28： For the RC circuit shown below, find its system function, draw its zero-pole plot, find its frequency response, and roughly sketch its frequency response curve.



Solution： The system function of the circuit is

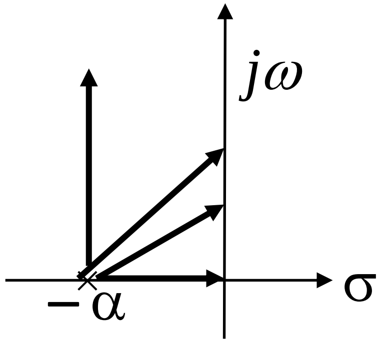


$$H(s)=\frac{U_o(s)}{U_i(s)}=\frac{\frac{1}{sC}}{R+\frac{1}{sC}}=\frac{1}{RCs+1}$$

$$=\frac{\frac{1}{RC}}{s+\frac{1}{RC}}=\frac{\alpha}{s+\alpha}\qquad\left(\alpha=\frac{1}{RC}\right)$$

4.9 Frequency Response from System Function Zero-pole Distribution

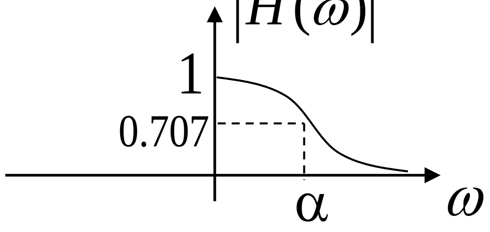
Roughly sketch the frequency response from $H(s)=\dfrac{\alpha}{s+\alpha}$:



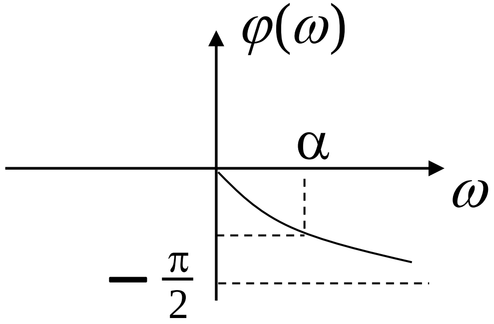
(1) when $\omega=0$, the pole vector points to the origin, with magnitude α and phase 0; thus, $|H(\omega)|=\alpha/\alpha=1$, $\varphi(\omega)=0$

(2) when $\omega\uparrow$, the magnitude of the pole vector↑, phase↑; $|H(\omega)|\downarrow$, $\varphi(\omega)\downarrow$.

(3) when $\omega=\alpha$, $|H(\omega)|\approx 0.707$, $\varphi(\omega)=-\pi/4$.



(4) when $\omega\to\infty$, $|H(\omega)|\to 0$, $\varphi(\omega)\to -\pi/2$.

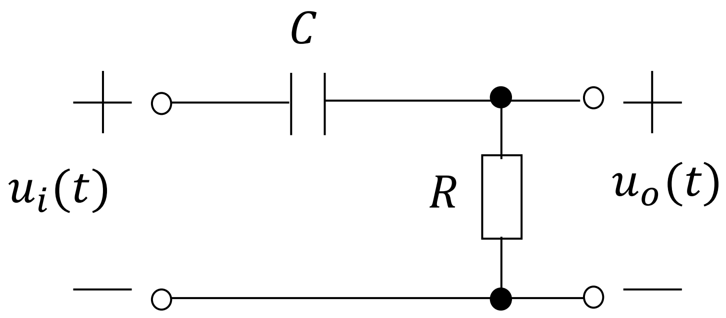


From its magnitude-frequency response curve, it can be seen that it is a **low-pass filter**.

$$H(\omega)=\frac{\alpha}{j\omega+\alpha}=\frac{\alpha}{\sqrt{\omega^2+\alpha^2}}e^{-j\,arctg(\frac{\omega}{\alpha})}$$

单选题 1分

For the RC circuit shown below, it represents a （ ）.



A Low-pass filter

B High-pass filter

C Band-pass filter

D All-pass filter

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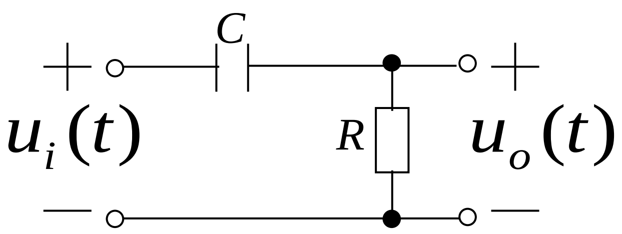
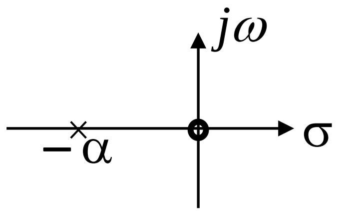
Example 4-29: For the RC circuit shown in the figure, find its system function, draw its zero-pole plot, find its frequency response, and roughly sketch its frequency response curve.



Solution: The system function of the circuit is derived as follows:

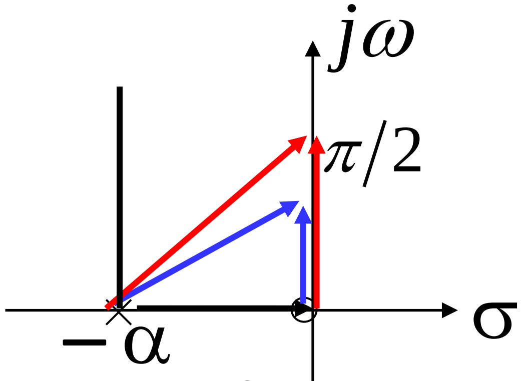
$$H(s)=\frac{U_o(s)}{U_i(s)}=\frac{R}{R+\frac{1}{sC}}=\frac{RCs}{RCs+1}$$

$$=\frac{s}{s+\frac{1}{RC}}=\frac{s}{s+\alpha}\qquad\left(\alpha=\frac{1}{RC}\right)$$



$$H(\omega)=\frac{j\omega}{j\omega+\alpha}$$

Roughly sketch the frequency response from $H(s)=\dfrac{s}{s+\alpha}$:



(1) when $\omega=0$， the pole vector points to the origin, with a magnitude of $\boldsymbol{\alpha}$ and a phase of **0;** the zero vector has a magnitude of 0 and a phase of $\boldsymbol{\pi/2}$; thus, $|H(\omega)|=0$, $\varphi(\omega)=\pi/2$.

(2) when $\boldsymbol{\omega}\uparrow$， the magnitude of the pole vector ↑， the phase↑; the magnitude of the zero vector ↑， **and its increasing rate is higher than that of the pole vector,** with a phase of $\boldsymbol{\pi/2}$; $|H(\omega)|\uparrow, \varphi(\omega)\downarrow=(\pi/2)-arctg(\omega/\alpha)$

For example， when the phase is $\boldsymbol{\pi/6}$, the magnitude of the zero vector is $\sqrt{3}\alpha/3$, the magnitude of the pole vector is $2\sqrt{3}\alpha/3$, and $|H(\omega)|=1/2$; when the phase approaches **π/4**, the magnitude of the zero vector approaches $\boldsymbol{\alpha}$， the magnitude of the pole vector approaches $\sqrt{2}\alpha$， and $|H(\omega)|\rightarrow 0.707$.

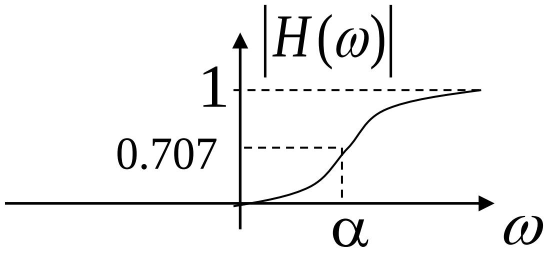
(3) when $\omega=\alpha$, $|H(\omega)|\approx 0.707$, $\varphi(\omega)=\pi/4$.

(4) when $\omega\rightarrow\infty$， both the zero vector and the pole vector are perpendicular to the real axis and point to infinity, with infinite magnitude and a phase of **π/2.**

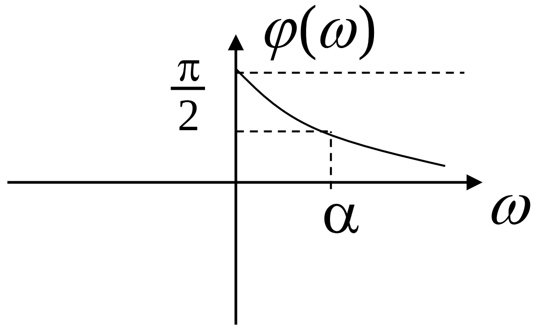
At this point, $|H(\omega)|\rightarrow 1$, $\varphi(\omega)\rightarrow 0$.

4.9 Frequency Response from System Function Zero-pole Distribution

$$|H(\omega)| = \frac{|\omega|}{\sqrt{\omega^2+\alpha^2}}$$



$$\varphi(\omega) = \begin{cases} \frac{\pi}{2} - arctg(\frac{\omega}{\alpha}) & \omega > 0 \\ -\frac{\pi}{2} - arctg(\frac{\omega}{\alpha}) & \omega < 0 \end{cases}$$



From its magnitude-frequency response curve, it can be seen that this is a **high-pass filter**.

$$H_{HP}(\omega) = \frac{j\omega}{j\omega+\alpha}, \quad H_{LP}(\omega) = \frac{\alpha}{j\omega+\alpha}$$

$$H_{HP}(\omega) = 1 - H_{LP}(\omega)$$

Homework

- Repeat the examples in the lecture notes.
- 6.47(b), 6.48(b) and 6.53(a) in the textbook.